

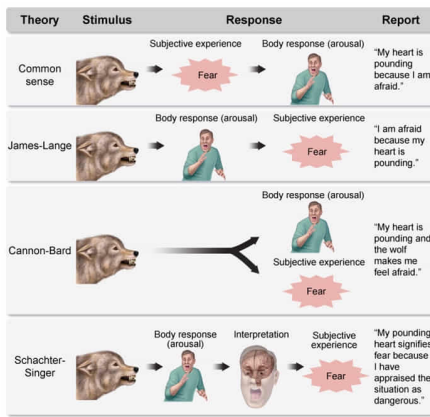
A researcher recruited 20 individuals with complete spinal cord injuries at cervical vertebrae C3 or C4 (participants had no sensory or motor function below the site of injury) and 20 matched controls. Participants were presented with images designed to evoke positive, negative, or no emotion. If the results indicate no significant difference between participants with spinal cord injuries and control participants in the emotional response to the images, this finding would:

- ✓ ☒ A. contradict the James-Lange theory. [43%]
☐ B. support the Schachter-Singer theory. [14%]
☐ C. contradict the Cannon-Bard theory. [17%]
☐ D. support the theory of universal emotions. [24%]

Correct

43% answered correctly

Explanation:



The **James-Lange theory** holds that specific **emotions** are elicited by stimuli (eg, loud noise) that produce **specific physiological reactions** (eg, increased heart rate), which are transmitted as sensory information to the brain via the spinal cord.

As a component of the **central nervous system (CNS)**, the spinal cord contains gray matter (nonmyelinated neuronal cell bodies, dendrites, axons) surrounded by white matter (primarily myelinated axons). Spinal cord **white matter** is composed of **afferent** (ascending) axonal tracts that carry **sensory information** from the body **to the brain** and efferent (descending) axonal tracts that carry motor commands from the brain to the body.

Individuals with **cervical** spinal cord injury (SCI) do not receive sensory information from the body. According to the James-Lange theory, such individuals are unable to experience emotion because sensory information (eg, increased heart rate) from the body cannot reach the brain; as such, this would predict that only control participants in the present scenario would be able to experience emotion.

Therefore, a finding that there is **no significant difference** between the **emotional responses** of **controls** and those with **complete SCI** **contradicts the James-Lange theory** by showing that controls and affected individuals display similar emotion.

(Choice B) According to the Schachter-Singer theory, emotion results from physiological arousal followed by cognitive appraisal (ie, sensory information *and* interpretation of environmental cues are required). Finding no significant difference between the emotional responses of controls and those with complete SCI would *contradict* (not support) the Schachter-Singer theory.

(Choice C) According to the Cannon-Bard theory, physiological arousal and emotion are independent processes that occur simultaneously (ie, the experience of emotion is not tied to sensory information from the body). Finding no difference between individuals with complete SCI and controls in the display of emotion would *support* (not contradict) the Cannon-Bard theory.

(Choice D) According to the theory of universal emotions, **certain emotions** are expressed and detected by everyone, regardless of culture. This study did not assess the emotional reaction of participants from diverse cultural backgrounds, so the results neither support nor contradict this theory.

Educational objective:

The James-Lange theory suggests that specific emotions result from stimuli eliciting specific physiological reactions, which are transmitted to the brain via the spinal cord white matter (afferent axonal tracts, which carry sensory information from the body to the brain).

Time spent:0 sec**Subject:**
Behavioral Sciences**QID:**402542**System:**
Motivation, Emotion, Attitudes, Personality, and Stress